

# The Chinese University of Hong Kong, Shenzhen

## Course Outline

### 1. Course identity

#### A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Course code            | STA5011                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Course title (English) | Multivariate Statistical Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Course title (Chinese) | 多元统计分析                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Units                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Description (English)  | This course deals with multivariate statistical analysis and their applications in the real world. Students will examine a variety of statistical methods for multivariate data, including multivariate distributions and inferences, analysis of variance, multivariate regression, dimension reduction techniques such as principal component analysis, factor analysis, canonical correlation analysis, discrimination and classification, and clustering methods. The integrated use of these techniques in analyzing real problems will be emphasized throughout the course. |
| Description (Chinese)  | 本课程讲授多元统计分析方法及其在现实世界中的应用。学生将学习多种多元数据分析的统计方法，包括多元分布与推断、方差分析、多元回归、降维技术（如主成分分析）、因子分析、典型相关分析、判别与分类方法以及聚类方法。课程将重点强调这些技术在分析实际问题中的综合应用。                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

#### B. Corresponding course at CUHK

Please give details of the closest corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

|                        |  |
|------------------------|--|
| Course code            |  |
| Course title (English) |  |
| Course title (Chinese) |  |
| Units                  |  |
| Description (English)  |  |
| Description (Chinese)  |  |

## **2. Course requisites**

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

### **A. Prerequisites**

NA

### **B. Co-requisites**

NA

### **C. Exclusion**

### **D. Others**

## **3. Learning outcomes**

Upon successful completion of this course, students should be able to

Knowledge:

- a) Understand the structure of a mean vector for one population or more than one population: tests on means and covariances, multivariate ANOVA
- b) Understand the linear associations among a group of variables: multiple, partial and canonical correlation; multivariate regression.
- c) Understand the latent linear structure of multiple measurements: dimension reduction, principal components analysis and factor analysis
- d) Understand the classification problem of a population with one or more than one measurement: discriminant analysis and classification.

Skills Generic:

- e) Identify the need for multivariate statistical techniques
- f) Recognize the appropriate multivariate method for a problem
- g) Apply the hands-on skills of analyzing multivariate data with statistical software R/Python
- h) Interpret results and make statistical inference

## **4. Course syllabus**

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

- 1) Overview and matrix orientation
- 2) Multivariate normal and related distributions
- 3) Inference about mean structure
- 4) Multivariate analysis of variance (MANOVA)
- 5) Multivariate regression and linear model
- 6) Tests of covariance and correlation matrices
- 7) Dimensionality reduction
- 8) Principal component analysis (PCA)
- 9) Factor analysis
- 10) Canonical correlation analysis (CCA)
- 11) Discrimination and classification

## 5. Assessment scheme

| Component/ method                            | % weight |
|----------------------------------------------|----------|
| Mid-term exam (1.5-hour written examination) | 20       |
| Final exam (2-hour written examination)      | 30       |
| Assignments                                  | 50       |

Remarks:

- 1) Students have to hand in assignments by the due dates.
- 2) Late submission of assignments will receive a score of ZERO.
- 3) Copied assignments (partially or wholly) will be penalized and/or reported as plagiarism.
- 4) AI tools (all tools on the University's AI platform) are permitted for use in coding work within the assignments. If AI tools are used for data analysis and interpretation, students should acknowledge AI and provide their own arguments/judgements regarding the results of AI tools.
- 5) AI tools are NOT allowed in the mid-term and final exams.

## 6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type: **General**

| Grade | Description                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A     | Demonstrates the ability to synthesize and apply the principles or subject matter learnt in the course, to novel situations and/or in novel ways, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research.<br>Has the ability to express the synthesis of ideas or application in a clear and cogent manner. |

|    |                                                                                                                                                                                                                                                                                                                        |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A- | Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive.<br>Has the ability to express the knowledge or application with clarity.                                                          |
| B  | Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive.<br>Has the ability to express the knowledge or application in a satisfactory and unambiguous way. |
| C  | Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented.<br>Has the ability to express the separate pieces of knowledge in an unambiguous way.           |
| D  | Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials<br>Has the ability to state the knowledge or application in simple terms.                                     |
| F  | Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.                                                                                                                                                                                                     |

## 7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

1. A course questionnaire answered by every student after finishing all the lectures
2. Feedback from conversations with students after class and office hour

## 8. Readings

### Textbook:

1. Applied Multivariate Statistical Analysis, 0131877151, Johnson, R.A. and Wichern, D.W., 6TH, United States

### Others:

## 9. Course components

| Activity    | Hours/week |
|-------------|------------|
| Assignments | 6          |
| Lectures    | 3          |

|          |   |
|----------|---|
| Tutorial | 1 |
|----------|---|

## 10. Indicative teaching plan

| Week | Content/ topic/ activity                           |
|------|----------------------------------------------------|
| 1    | Overview and introduction                          |
| 2    | Matrix algebra                                     |
| 3    | Multivariate normal and related distributions (I)  |
| 4    | Multivariate normal and related distributions (II) |
| 5    | Inferences regarding a mean vector                 |
| 6    | Multivariate analysis of variance (MANOVA)         |
| 7    | Multivariate regression and linear models          |
| 8    | Midterm Examination                                |
| 9    | Tests of covariance and correlation matrices       |
| 10   | Correlation analysis                               |
| 11   | Dimensionality reduction                           |
| 12   | Principal components analysis (PCA)                |
| 13   | Factor analysis                                    |
| 14   | Discriminant analysis and classification           |

## 11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

60 students for lecture (x 1); 30 students for tutorials (x 2)

## 12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

## 13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

提交申请

**14. Any other information**

**15. Version date**

|                |            |
|----------------|------------|
| Version number | 1          |
| As of (date)   | 2025-11-02 |